1 A car of mass m is travelling along a straight, horizontal road at a constant speed v.

At time $t = 0$, the driver of the car sees an obstruction in the road ahead of the car and applies the brakes.

The car does **not** begin to decelerate at $t = 0$.

(a) Explain what is meant by deceleration.

.....

.....

..... [2]

(b) Suggest **one** reason why the car does **not** begin to decelerate at $t = 0$.

.....

..... [1]

(c) Fig. 1.1 is the distance–time graph for the car from $t = 0$.

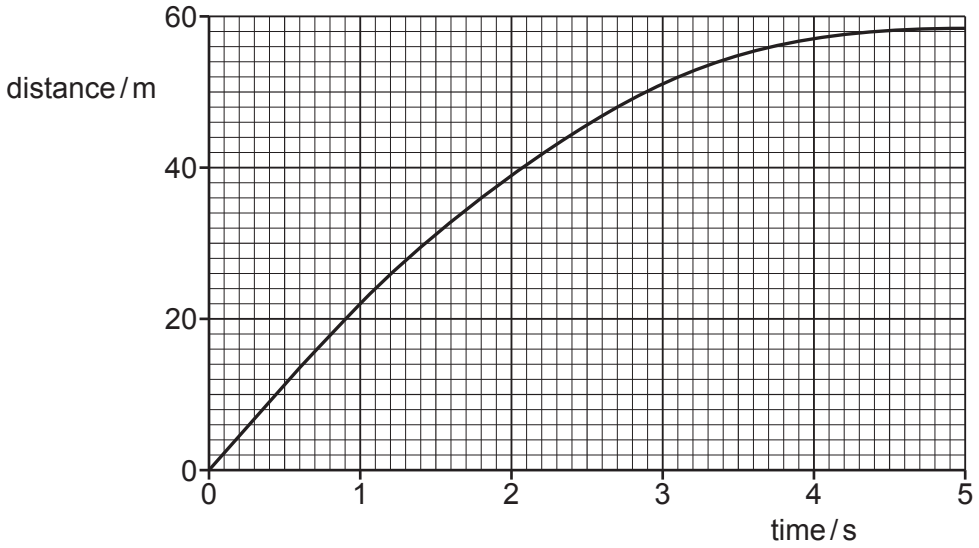


Fig. 1.1

(i) State the property of a distance–time graph that corresponds to speed.

..... [1]

(ii) Using Fig. 1.1, determine the initial speed v of the car.

v = ... [2]

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(d) When the car is decelerating, there is a constant resistive force F on the car due to the brakes.

The deceleration of the car is greater than $\frac{F}{m}$ and is **not** constant.

Explain why:

(i) the deceleration of the car is greater than $\frac{F}{m}$

.....

..... [1]

(ii) the deceleration is **not** constant.

.....

.....

..... [2]

[Total: 9]